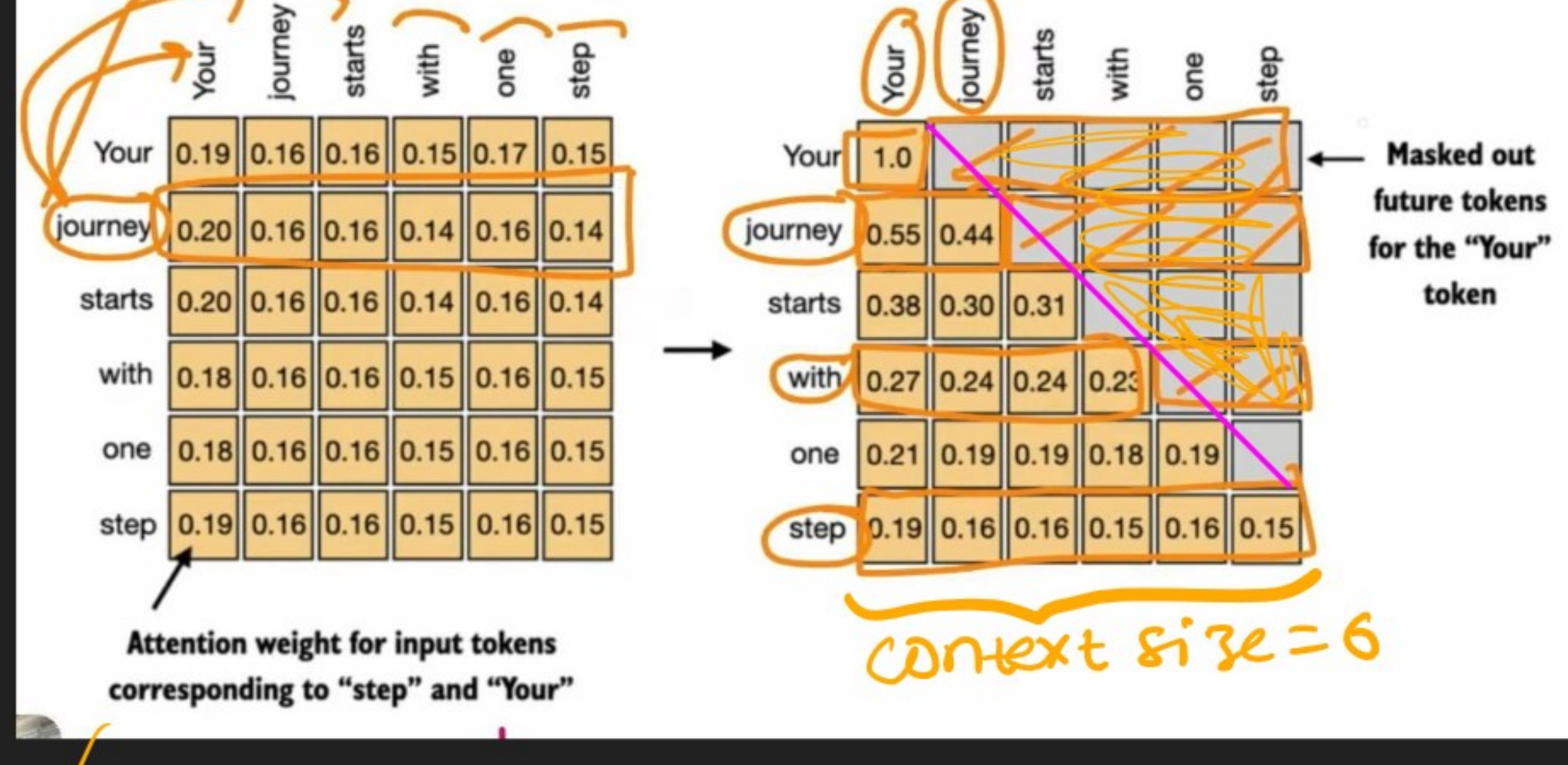


lecture - 16 (causal Attention)

- ① causal attention, also known as masked attention is a special form of self-attention.
- ② It restricts the model to only consider previous and current inputs in a sequence when processing any given token.
- ③ This is in contrast to the self-attention mechanism, which allows access to the entire input sequence at once
- ④ when computing attention scores, the causal attention mechanism that the model only factors in tokens that occur at or before the current token in the sequence
- ⑤ To achieve this in GPT like LLMs, for each token passed, we mask out the further tokens, which comes after the current token in the input text

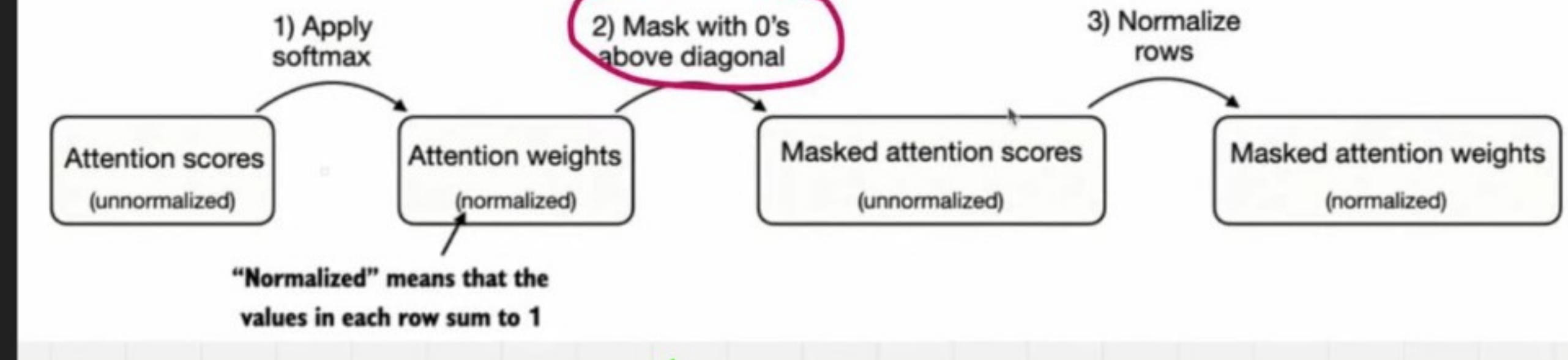


we masked out the attention weight above the diagonal, and we normalize the non masked attention weights, such that the attention weight sum up to 1 in each row

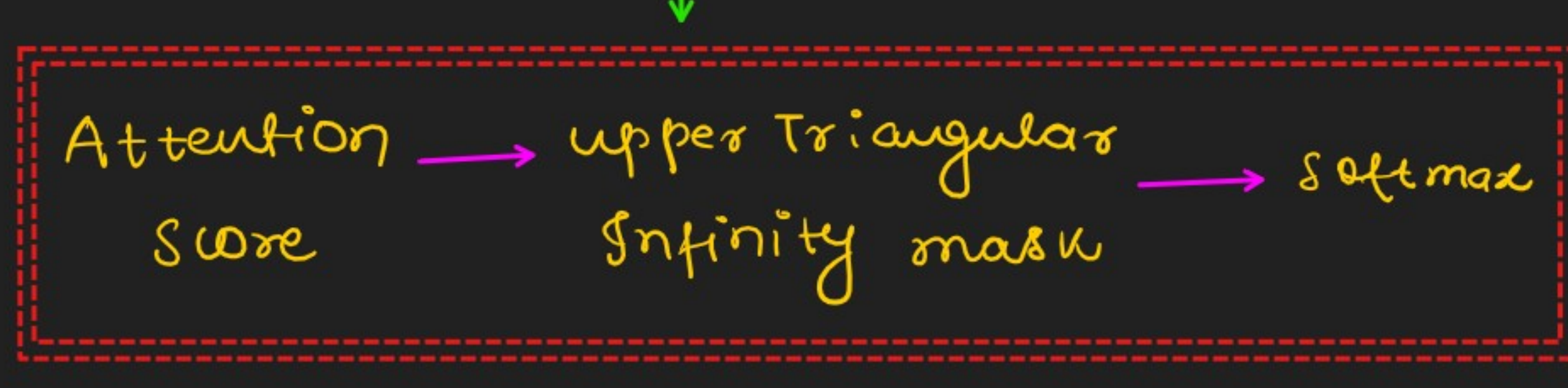
* Applying a causal Attention Mask *

Strategy : Get attention weights

→ Zero out element above the diagonal and normalize the resulting matrix.



more efficient way



* Masking additional attention

weight dropout *

Dropout is a deep learning technique where randomly selected hidden layer unit are ignored during training

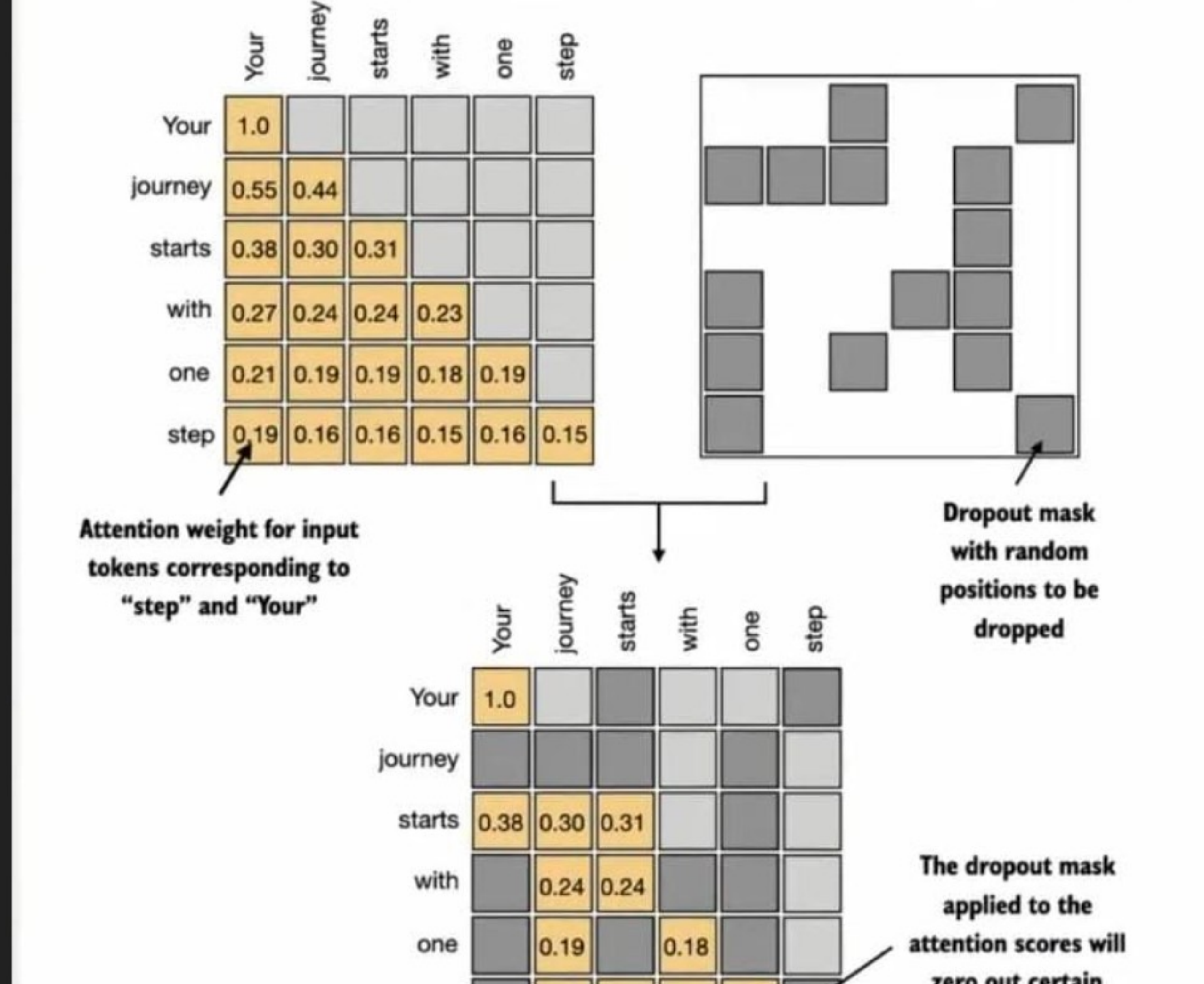
→ This prevents overfittings and improves generalization performance.

→ In transformer architecture, including model like GPT, dropout in the attention mechanism is applied in 2 specific areas.

after calculating attention scores

after applying attention weight to value vector

→ Applying dropout after calculating attention weights is more common, and we consider that



* Now, we can implement a causal Attention class, which incorporates causal attention and Dropout modification into the self-attention class we implemented earlier.